EXNO: 5

30/4/2026

EXPERIMENT ON PACKET CAPTURE TOOL: WIRESHARK

AIM:

To understand the feature of wireshark as a Packet capture tool and understand encapsulation of information at various layers of a protocol stack.

PACKET SNIFFER:

-> Capture packets sent and received by the computer
-> Displays protocols information of captured packets
-> works passively
-> Receives copies of networks packets for analysis.

PACKET SNIFFER TOOLS:

-> TCP dump - Command line packet capture tool.
-> wireshark - Graphical packet analyzer.

DESCRIPTION:

Wireshark is a Networks protocols analyzer that captures and displays networks packets in real time. It helps monitor, analyze, troubleshoot and inspect network traffic using filters and packet details.

dns

No.	Time	Source	Destination	Protocol	Lengtl	Info
5047	64.823837	172.16.75.31	172.16.72.1	DNS	70	Standard query 0xf2bf HTTPS dns.google
5048	64.823968	172.16.75.31	172.16.72.1	DNS	70	Standard query 0x2c40 A dns.google
5092	64.872708	172.16.72.1	172.16.75.31	DNS	102	Standard query response 0x2c40 A dns.go
5093	64.872708	172.16.72.1	172.16.75.31	DNS	70	Standard query response 0xf2bf HTTPS dn
5907	73.507194	172.16.75.31	172.16.72.1	DNS	70	Standard query 0xdd84 HTTPS dns.google
5908	73.507610	172.16.75.31	172.16.72.1	DNS	70	Standard query 0x7b95 A dns.google
5909	73.522819	172.16.72.1	172.16.75.31	DNS	102	Standard query response 0x7b95 A dns.go
5910	73.522819	172.16.72.1	172.16.75.31	DNS	70	Standard query response 0xdd84 HTTPS dn
6172	75.347451	172.16.75.31	8.8.8.8	DNS	70	Standard query 0x7402 HTTPS dns.google
6173	75.347571	172.16.75.31	8.8.8.8	DNS	70	Standard query 0xddfb A dns.google
6174	75.377779	8.8.8.8	172.16.75.31	DNS	102	Standard query response 0xddfb A dns.go
6180	75.398557	8.8.8.8	172.16.75.31	DNS	146	Standard query response 0x7402 HTTPS dn

```
> Frame 5047: Packet, 70 bytes on wire (560 bits), 70 bytes
> Ethernet II, Src: CloudNetwork_78:00:0b (4c:82:a9:78:00:0
> Internet Protocol Version 4, Src: 172.16.75.31, Dst: 172.
> User Datagram Protocol, Src Port: 53918, Dst Port: 53
> Domain Name System (query)
```

```
0000  7c 5a 1c cf be 41 4c 82  a9 78 00 0b 08 00 45 00  |
0010  00 38 65 85 00 00 80 11  e9 ee ac 10 4b 1f ac 10  8
0020  48 01 d2 9e 00 35 00 24  99 73 f2 bf 01 00 00 01  H
0030  00 00 00 00 00 00 03 64  6e 73 06 67 6f 6f 67 6c  .
0040  65 00 00 41 00 01                                e
```

arp

No.	Time	Source	Destination	Protocol	Lengtl	Info
1343	10.957901	CloudNetwork_78:90:...	Broadcast	ARP	42	Who has 172.16.75.125? Tell 172.16.75.6
1977	14.358615	CloudNetwork_78:90:...	Broadcast	ARP	42	Who has 172.16.77.18? Tell 172.16.75.61
2098	15.471882	CloudNetwork_78:90:...	Broadcast	ARP	42	Who has 172.16.77.18? Tell 172.16.75.61
2400	17.952584	CloudNetwork_78:90:...	Broadcast	ARP	42	Who has 172.16.78.5? Tell 172.16.75.61
2747	24.883561	CloudNetwork_78:90:...	Broadcast	ARP	42	Who has 172.16.75.204? Tell 172.16.75.6
4906	61.332363	CloudNetwork_77:ff:...	Broadcast	ARP	42	Who has 172.16.77.93? Tell 172.16.75.53

USES OF NETWORKS:

=> Captures networks traffic
=> Analyzes protocols
=> Applies displays filters
=> Monitors Networks issues.

APPLICATION:

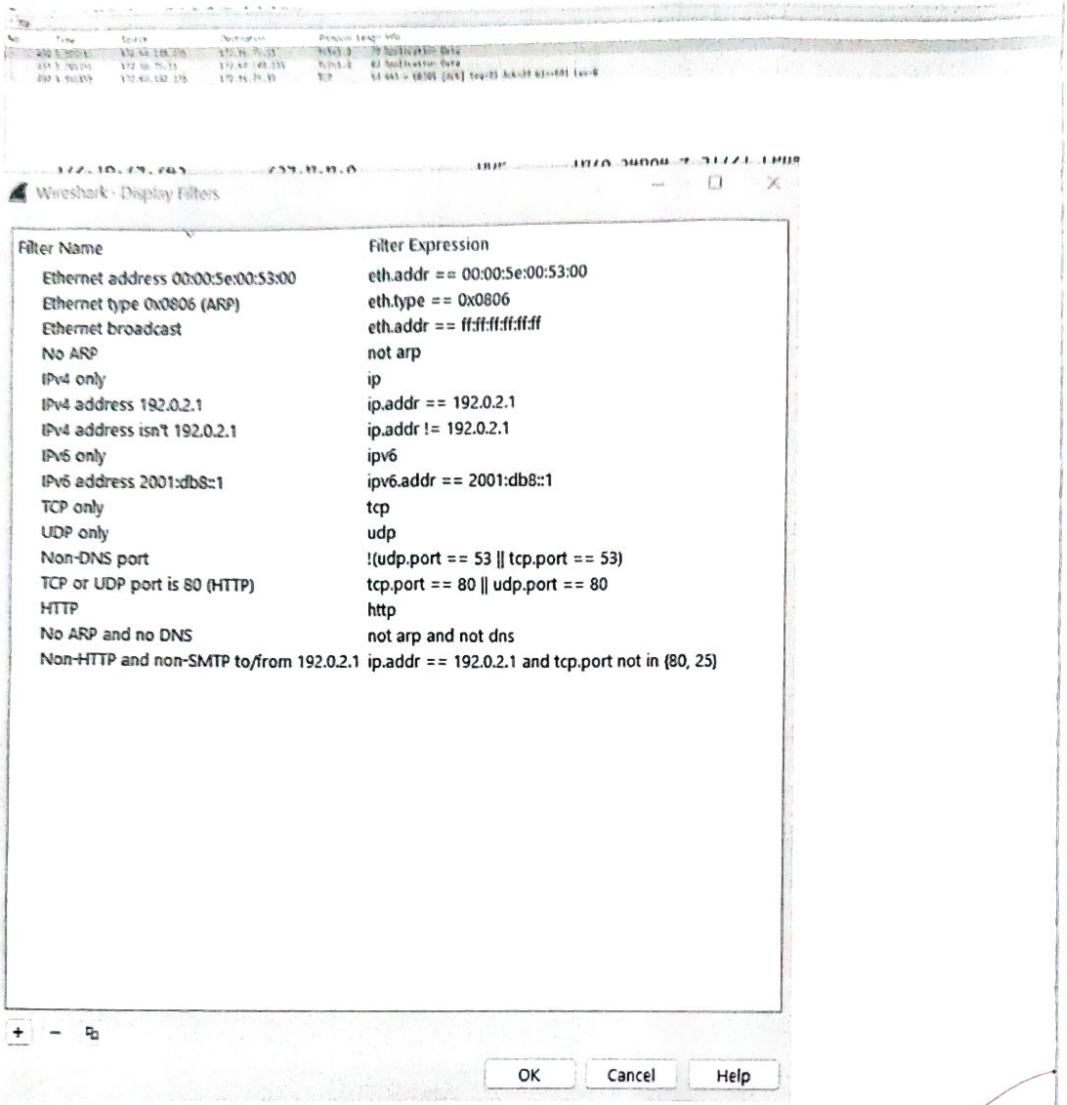
=> Network troubleshooting
=> Security Analysis
=> Protocol debugging
=> Learning network protocols

CAPTURING PACKET:

1. open wireshark
2. Select the network interface
3. Click start captures
4. Observe packet in real time
5. Click stop and save the captures.

COLOR CODING:

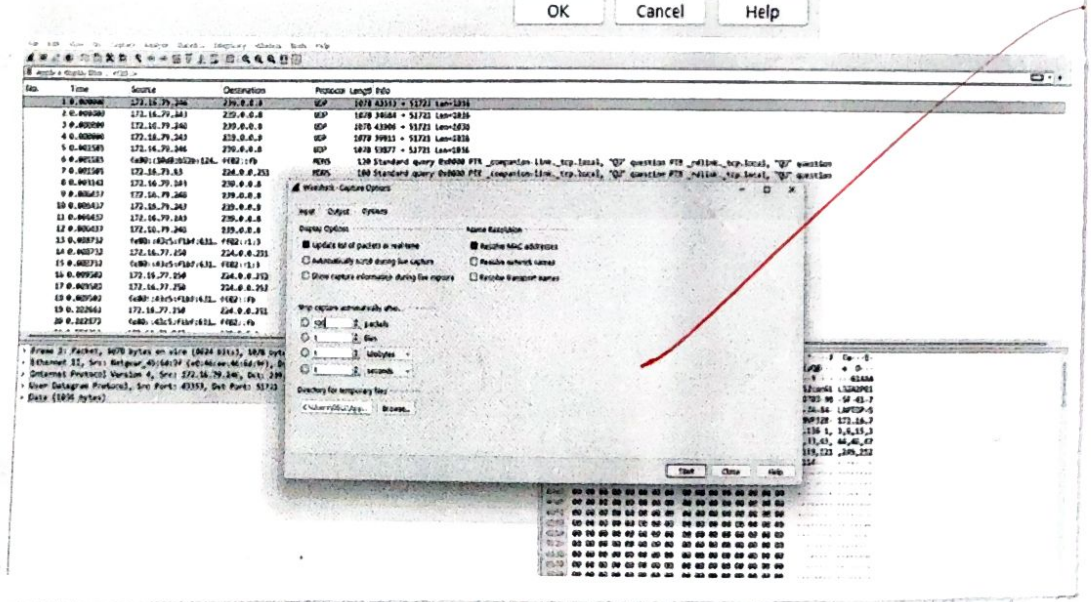
=> Purple : TCP
=> Blue : UDP
=> Black : Error Packets



Wireshark · Display Filters

Filter Name	Filter Expression
Ethernet address 00:00:5e:00:53:00	eth.addr == 00:00:5e:00:53:00
Ethernet type 0x0806 (ARP)	eth.type == 0x0806
Ethernet broadcast	eth.addr == ff:ff:ff:ff:ff:ff
No ARP	not arp
IPv4 only	ip
IPv4 address 192.0.2.1	ip.addr == 192.0.2.1
IPv4 address isn't 192.0.2.1	ip.addr != 192.0.2.1
IPv6 only	ipv6
IPv6 address 2001:db8::1	ipv6.addr == 2001:db8::1
TCP only	tcp
UDP only	udp
Non-DNS port	!(udp.port == 53 || tcp.port == 53)
TCP or UDP port is 80 (HTTP)	tcp.port == 80 || udp.port == 80
HTTP	http
No ARP and no DNS	not arp and not dns
Non-HTTP and non-SMTP to/from 192.0.2.1	ip.addr == 192.0.2.1 and tcp.port not in {80, 25}

OK Cancel Help



FILTERING PACKETS:

- Enter a filter in the filter box.
- Right click to apply filters or follow the TCP stream.

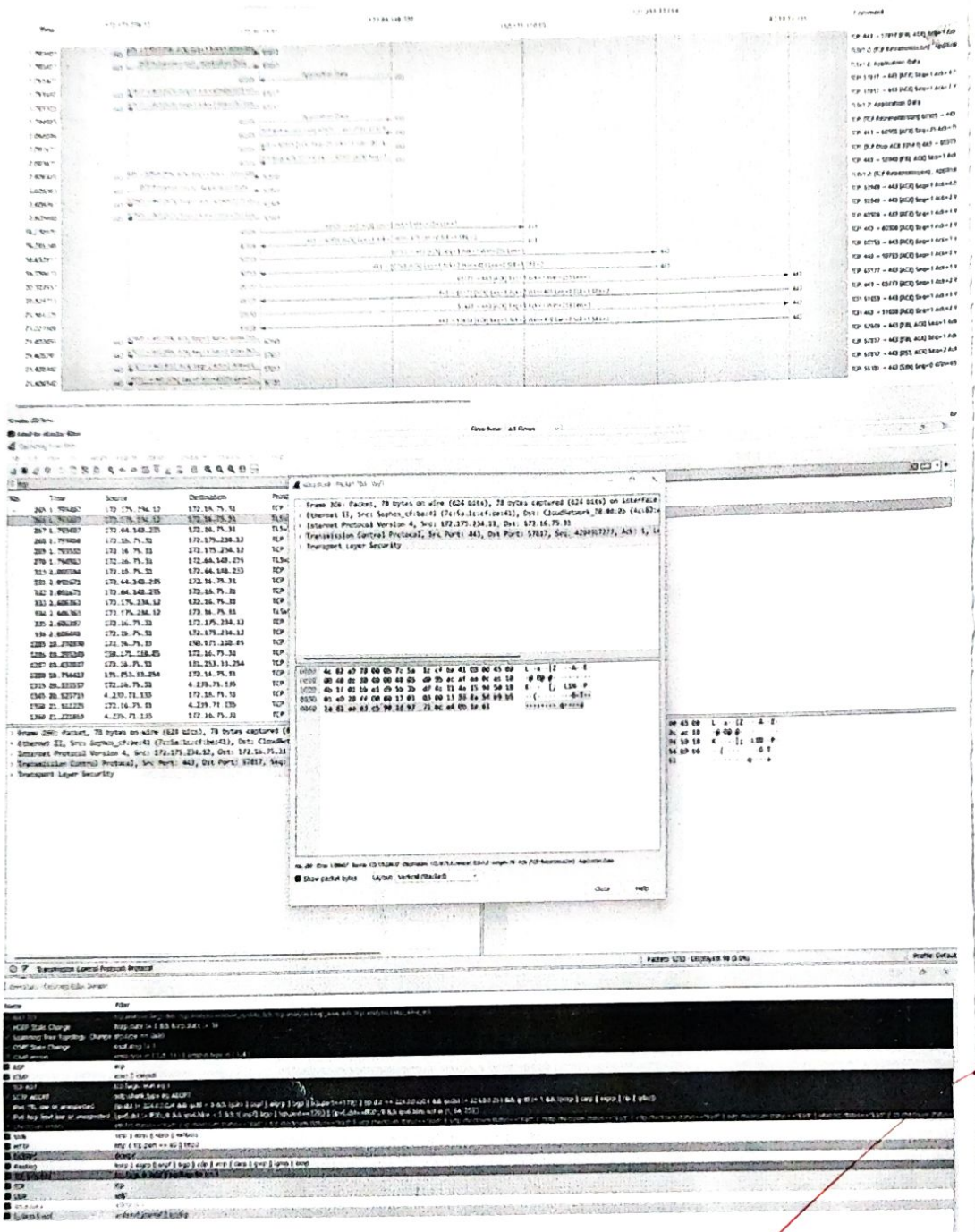
FLOW GRAPH:

Displays the communication flow between devices making packets exchange easier to understand.

PROCEDURE:

Analyzing Packets using Wireshark tool.

1. Create a filter to Displays only TCP/UDP packets.
   1. Open the capture file
   2. Apply the TCP || UDP filter
   3. Inspect and view the flow graph
2. Create a file to display only ARP Packets
   1. Open the capture file
   2. Apply the arp filter
   3. Inspect and view the flow graph.



3. Create a filter to display only DNS packets

1. Open the capture file
2. Apply the DNS filter
3. Inspect and view the flow graph

4. Create a filter to display only IP/ICMP packets

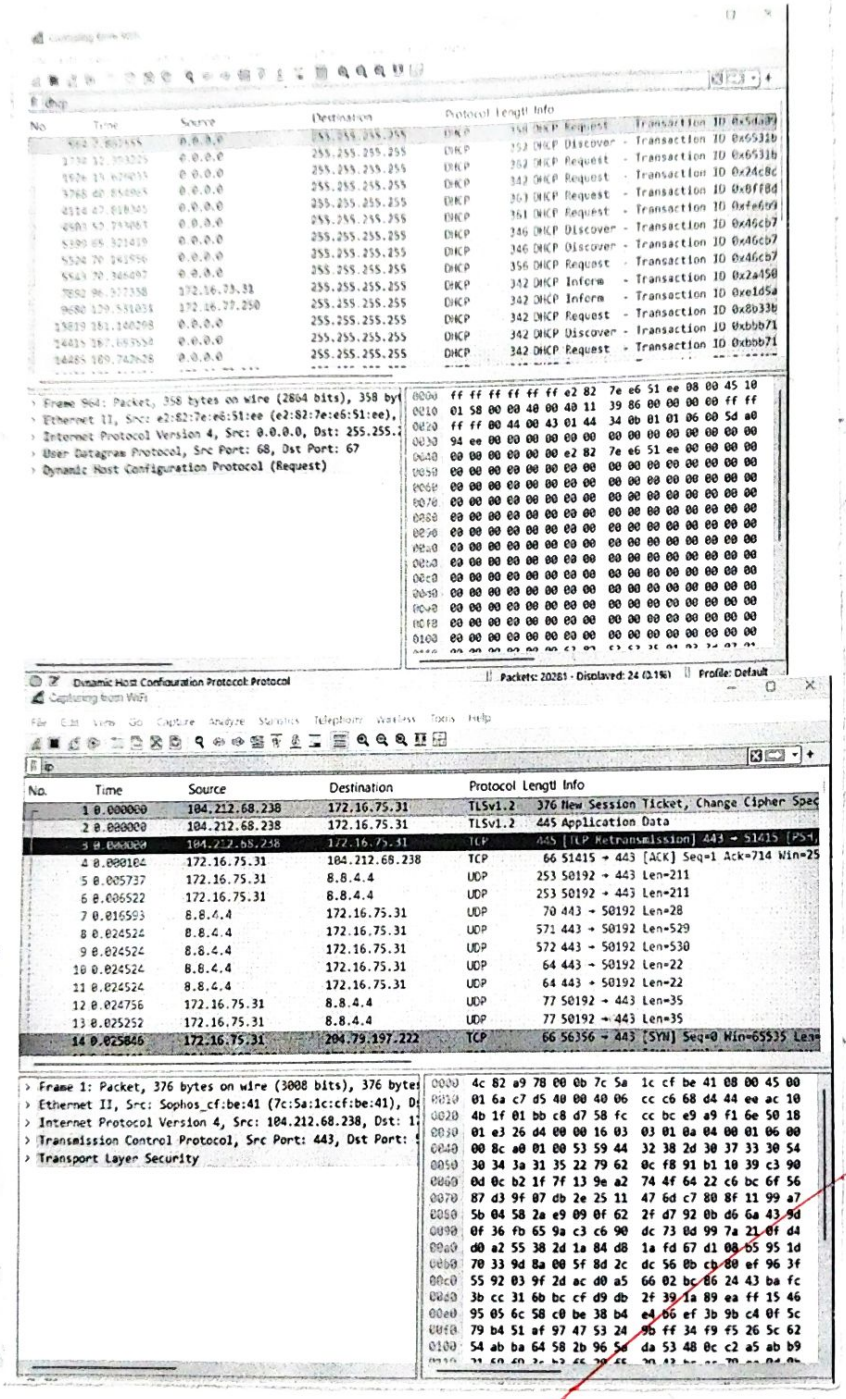
1. Open the capture file
2. Apply the IP or ICMP filter
3. Inspect and view the flow graph and ~~save~~ save captures.

5. Create a filter to display only DHCP packet

1. Open the capture file
2. Apply the DHCP file
3. Inspect and view the flow graph and save the capture

RESULT:

Thus the feature of wireshark and the implement of different filter was successfully completed.


dhcp
No. Time Source Destination Protocol Length Info
DHCP Request - Transaction ID
DHCP Discover - Transaction ID
DHCP Inform - Transaction ID
Frame 964: Packet, 358 bytes on wire (2864 bits), 358 by
Ethernet II, Src: e2:82:7e:e6:51:ee (e2:82:7e:e6:51:ee),
Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.
User Datagram Protocol, Src Port: 68, Dst Port: 67
Dynamic Host Configuration Protocol (Request)
Dynamic Host Configuration Protocol Protocol
Capturing from WiFi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
No. Time Source Destination Protocol Length Info
1 0.000000 104.212.68.238 172.16.75.31 TLSv1.2 376 New Session Ticket, Change Cipher Spec
2 0.000000 104.212.68.238 172.16.75.31 TLSv1.2 445 Application Data
3 0.000000 104.212.68.238 172.16.75.31 TCP 445 [TLP Retransmission] 443 → 51415 [PSH,
4 0.000104 172.16.75.31 104.212.68.238 TCP 66 51415 → 443 [ACK] Seq=1 Ack=714 Win=25
5 0.005737 172.16.75.31 8.8.4.4 UDP 253 50192 → 443 Len=211
6 0.006522 172.16.75.31 8.8.4.4 UDP 253 50192 → 443 Len=211
7 0.016593 8.8.4.4 172.16.75.31 UDP 70 443 → 50192 Len=28
8 0.024524 8.8.4.4 172.16.75.31 UDP 571 443 → 50192 Len=529
9 0.024524 8.8.4.4 172.16.75.31 UDP 572 443 → 50192 Len=530
10 0.024524 8.8.4.4 172.16.75.31 UDP 64 443 → 50192 Len=22
11 0.024524 8.8.4.4 172.16.75.31 UDP 64 443 → 50192 Len=22
12 0.024756 172.16.75.31 8.8.4.4 UDP 77 50192 → 443 Len=35
13 0.025252 172.16.75.31 8.8.4.4 UDP 77 50192 → 443 Len=35
14 0.025846 172.16.75.31 204.79.197.222 TCP 66 56356 → 443 [SYN] Seq=0 Win=65535 Len=
Frame 1: Packet, 376 bytes on wire (3008 bits), 376 bytes
Ethernet II, Src: Sophos_cf:be:41 (7c:5a:1c:cf:be:41), D
Internet Protocol Version 4, Src: 104.212.68.238, Dst: 1
Transmission Control Protocol, Src Port: 443, Dst Port:
Transport Layer Security